Dataset of the high-frequency and ultrasound personal exposimeter (HiFUSPEx) validation scans performed in three simulated industrial sound fields.

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Abstract

All the measurement data contained in this dataset were collected during the performance validation process of the newly developed high-frequency and ultrasound personal exposimeter (HiFUSPEx). The validation process of the HiFUSPEx was performed as a part of the collaborative project of the Physikalisch-Technische Bundesanstalt and the Institute for Occupational Safety and Health of the German Social Accident Insurance titled "Development of a high-frequency sound exposimeter for the occupational health protection". The validation process was designed as a comparative analysis of two-dimensional scans of various idealized ultrasonic sound fields. The ultrasonic sound fields were idealized in such a way, so that they resemble real-world industrial scenarios as closely as possible, and at the same time they provide sufficiently stable ultrasonic environment to conduct experiments over a prolonged period of time. Each simulated industrial scenario was in turn scanned with the HiFUSPEx and an ultrasound level meter, whose ultrasound detection performance was previously tested up to 100 kHz. In order to assess ultrasound measurement performance of the HiFUSPEx when exposed to varied sound fields, three pieces of industrial equipment were simulated: an ultrasonic welding machine, an ultrasonic cleaning bath, and a compressed-air gun. These were chosen as most representative of the ultrasonic equipment used at industrial workplaces in Germany. Ultrasonic signals, specific for each simulated scenario, were produced using a bespoke signal generation chain built for the purposes of this study. To further improve the resemblance to an industrial situation an artificial head was placed in front of the simulated industrial equipment. As a result, the influence of a worker within an ultrasonic field was accounted for. The two-dimensional scans were performed on a vertical surface with dimensions 32 × 32 cm (width × height), 10 cm away from the entrance to the ear canal with the lateral resolution of 10 mm. Each simulated scenario was surveyed with the artificial head's height adjusted in turn to sedentary and standing positions. The measurement data in the dataset were recorded as sound pressure over time using WAVE-file format. These can be computationally converted to sound pressure level (SPL) using the given calibration values for each device used. The high-resolution data being published here can also be used for other applications.

1 Introduction

The dataset consists of measurement data taken as a part of the collaborative project of the Physikalisch-Technische Bundesanstalt (PTB) and the Institute for Occupational Safety and Health of the German Social Accident Insurance (IFA of the DGUV) titled "Development of a high-frequency sound exposimeter for the occupational health protection". The data collected in this dataset were used for the performance validation investigation of the newly developed high-frequency and ultrasound personal exposimeter (HiFUSPEx), which was described in detail in reference [1]. Therein, the authors present and discuss a comparison of the measurement results of the HiFUSPEx and a previously developed and tested ultrasound level meter (USPM) [2].

The dataset published here can, for instance, provide a visual representation of example ultrasound fields. This delivers an important foundation for future analyses of similar fields, where adequate assessment and measuring techniques may be required. Moreover, datasets for high-resolution scans of ultrasound fields around a real industrial welding machine were previously published [3]. These can be used together with the data of idealized ultrasound fields published here to further appreciate the complexity of such sound fields.

This dataset is divided into three subsets each documenting the scan of a certain simulated scenario; these are: a cleaning bath, a compressed-air gun and a welding machine. Each experimental scenario was measured with the protection grid of the used microphone removed. Only in the case of the welding machine measurements were additionally performed with the protection grid mounted on the microphone cartridge. These, however, were only performed with the height of the artificial head adjusted to the standing position of the worker. Signals generated in each scenario were largely based on the examples presented in VDI 3766:2012 - 09 [4] with the exception of the welding machine, where three different working frequencies of the welding machine were simulated; these are: 10 kHz sine wave, 20 kHz sine wave, and 10 kHz + 20 kHz multi-sine wave.

Given that the ISO 9612:2009 [5] states that noise exposure in an industrial setting should be investigated in standing (155 cm above the floor level) and sedentary positions (80 cm above the seat level) of the worker, both positions were investigated in each simulated scenario. An artificial head simulating the industrial worker was placed in the center of a cubic anechoic chamber with the approximate dimensions of 2 m $\times$ 2 m $\times$ 2 m. Scans were performed in vertical direction from the left and the right side of the artificial worker. The area covered was a 32 $\times$ 32 cm square with the origin at the entrance to the ear canal of the artificial head. Lateral resolution of each scan was set to 10 mm. An example view from the left side of the artificial head is presented in Fig. 1. There were 26 scans performed in total for each measuring device. Each scan consisted of 1089 measurement points equaling in total to 28,548 points scanned.

2 Data organisation

This dataset is organized as a single *.zip file where the recorded data can be accessed by navigating through a sequence of subdirectories whose names describe the arrangement of a particular measurement setting. In the case of the ultrasonic cleaning bath or compressed-air gun the organization is defined as shown in Fig. 2. In the case of the welding machine the organization is very similar with additional two levels between the Industrial Equipment and the Position directories. These additional levels are presented in Fig. 3.

Directory **AnalysedData** contains the data for all the points scanned in the specific measurement arrangement as indicated through the names of subdirectories. There is one *.csv file for each measurement position. Each column with measurement data in the *.csv has its frequency and/or



Figure 1: An example view from the left side of the artificial head showing the scanning area marked by the black frame. Scanning axes together with the USPM microphone at the origin of the scans are also shown.

```
Industrial Equipment (CleaningBath or CompressedAirGun)
└─ Position (Sedentary or Standing)
   └─ Side (Left of Right)
      └─ Device (HiFUSPEx or USPM)
         └─ Type of data (AnalysedData or CalculatedFigureSPL or SelectedRawData)
```

Figure 2: Directory tree for the cases of the cleaning bath and the compressed-air gun.

```
Industrial Equipment (WeldingMachine)
└─ Protection Grid (WithoutProtectionGrid or WithProtectionGrid)
   └─ Signal Type (10kHz or 20kHz or 10+20kHz)
      └─ Position (Sedentary or Standing)
         └─ ...
```

Figure 3: Additional levels in the directory tree for the case of the welding machine.

time weighting specified in the title row (the first row in the *.csv file). In the case of third-octave-band filters the center frequency of the appropriate frequency band is given, e.g. FB 5 kHz. Further information related to frequency and time weightings implemented are presented in Table 1.

CalculatedFigureSPL contains a single *.csv file, where an average sound pressure level is calculated for all the measurement points in a single scan. The data is organized so that columns

correspond to Z-axis and rows to Y-axis values as depicted in Fig. 1. An example sound field produced from data contained in such a file is shown in Fig. 4.

Table 1: Description of frequency and time weightings implemented for the measurements performed with the HiFUSPEx.

Name	**Type**	**Comments**
A, C, Z	Frequency weighting	Implemented according to IEC 61672:2013 [6]
U, AU	Frequency weighting	Implemented according to IEC 61012:1998 [7]
F, S	Time weighting	Implemented according to IEC 61672:2013 [6]
PEAK	Greatest sound level	
EQ	Time-averaged sound level over the complete measurement time	
E	Sound exposure	
E8h	Sound exposure (E)	Projected over an 8-hour workday
BLOCK PEAK	Greatest sound level	Over the last measured block
BLOCK EQ	Time-averaged sound level	Over the last measured block

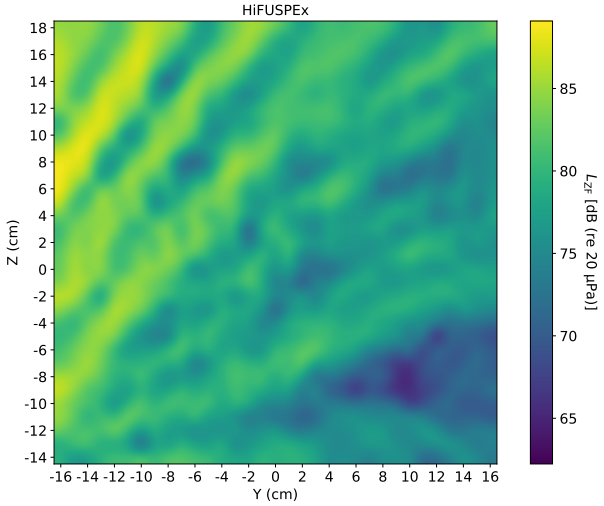


Figure 4: A example sound field as seen from the right side of the artificial head in the sedentary position.

The **SelectedRawData** directory contains raw and analyzed data for the selected number of points in each measurement arrangement. The calibration data necessary for the calculation of absolute sound pressure values from the raw data (WAVE-files) are included in this repository. In the case of the simulated welding machine these are: **bottom right point, center point, local**

maximum, local minimum and top left. For the case of the cleaning bath and the compressed-air gun these are: **bottom right point, center point, absolute maximum, top left**. These points were chosen because they represent the most significant areas of each sound field.

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Attached Files

- Calibration data of the HiFUSPEx ("HiFUSPEx_CalibrationData.txt")
- Calibration data of the USPM ("USPM_CalibrationData.txt")
- Description of dataset ("Dataset_Description.pdf")
- Dataset ("Dataset.zip")

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